OR-AS Operations Research Applications and Solutions	Case Name: **Garden Center**	Sector	Construction (commercial building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions
Date	**2015**	Project Control	Automatic tracking
File Name	**C2015-08 Garden Center**		Tracking based on user input

1. Project description

Project authenticity

A garden center construction project involving structural assembly, groundwork, roofing, utilities, interior and exterior finishing, roadwork, landscaping, and retail space setup across multiple phases.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	186
Planned Duration (PD)	191 days*
Budget At Completion (BAC)	467 297.2 €
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	14%
Activity Distribution (AD)	52%
Length of Arcs (LA)	0%
Topological Float (TF)	79%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	10.1	8.6	-2.1
CRI-rho	10.3	8.8	-1.9
CRI-tau	8.7	6.8	-1.4

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	14.9	35.1	-2.0
SI	3.4	11.0	-7.1
SSI	1.8	6.6	-6.8
CRI-r	8.8	8.2	-3.8
CRI-rho	8.7	8.2	-3.8
CRI-tau	9.0	7.0	-2.1

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	23.4	-8.7
PV - SPI	36.8	8.0
PV - SCI	37.0	8.0
ED - 1	28.5	-19.5
ED - SPI	36.8	8.0
ED - SCI	36.9	8.0
ES - 1	9.2	-8.2
ES - SPI(t)	24.3	24.1
ES - SCI(t)	24.3	24.1

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1.2	0.2
CPI	1.3	0.3
SPI	11.3	11.2
SPI(t)	11.5	11.5
SCI	11.3	11.2
SCI(t)	11.5	11.5
0.8 CPI + 0.2 SPI	6.6	6.3
0.8 CPI + 0.2 SPI(t)	5.7	5.5

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 8 tracking periods with a length of approximately 5 week(s). The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-4772.5	-4285.1	-61.3	1.0	1.0	1.0	0.9
std dev	5709.0	9255.7	52.6	0	0.1	0.1	0.1
final	5397.0	0	-72.0	1.0	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	191 days	Real Duration	200 days	Late	4.7%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	188.4	1.6	0.8	0.8
PV - SPI	189.1	1.7	0.7	0.7
PV - SCI	189.9	2.0	0.7	0.7
ED - 1	190.5	2.1	0.6	0.6
ED - SPI	191.1	2.0	0.5	0.5
ED - SCI	191.9	1.7	0.5	0.5
ES - 1	192.6	1.6	0.4	0.4
ES - SPI(t)	193.4	1.6	0.4	0.4
ES - SCI(t)	194.1	1.7	0.4	0.4

2.3.3.3. Cost forecasting

BAC	467 297.2 €	Real Cost	461 899.6 €	Under Budget	-1.2%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	472069.1	5708.9	0.4	-0.4
CPI	473312.6	6855.9	0.4	-0.4
SPI	467916.4	19934.8	0.4	-0.4
SPI(t)	475747.3	15544.1	0.4	-0.4
SCI	469207.5	20982.4	0.4	-0.4
SCI(t)	477141.6	17106.0	0.4	-0.4
0.8 CPI + 0.2 SPI	472107.0	8919.0	0.4	-0.4
0.8 CPI + 0.2 SPI(t)	473706.4	8298.6	0.4	-0.4